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Air filter and gasket for the same

Abstract

Disclosed herein are replaceable filter frame gaskets and air filters having the same. The gasket is secured elastically to a filter frame of the air filter, making the gasket easily reusable and removable from the air filter.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6117202	12/1999	Wetzel	N/A	N/A
7261757	12/2006	Duffy	N/A	N/A
9409112	12/2015	Wagner et al.	N/A	N/A
2003/0034279	12/2002	Baltzer et al.	N/A	N/A
2013/0269531	12/2012	Crabtree et al.	N/A	N/A
2013/0291502	12/2012	Gorman	N/A	N/A
2016/0303501	12/2015	Musone	N/A	N/A
2019/0344208	12/2018	Morgan et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104039420	12/2013	CN	N/A

OTHER PUBLICATIONS

Parker Hannifin Corporation 2019, QuadSEAL® Filters MERV 14. cited by applicant
Blue Heaven Technologies, Test Report ANSI/ASHRAE Standard 52.2-20117 Rev. 5-Feb. 27, 2017. cited by applicant
Parker, HVAC Filtration Division, Revision Date Dec. 17, 2018. cited by applicant
International Application PCT/US2021/038123, International Search Report and Written Opinion, dated Sep. 24, 2021, consists of 12 pages. cited by applicant
Brazilian Office Action for Application No. 1120220263910 dated Mar. 25, 2025. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/352,106, filed Jun. 18, 2021, which claims benefit of U.S. Provisional Application Ser. No. 63/042,407, filed Jun. 22, 2020, which is incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) Embodiments described herein relate to air handling systems utilizing air filters. More particularly, embodiments described herein relate to air filter gaskets for the same.

Description of the Related Art

(2) Air filtration systems typically utilize various combinations of pre- and primary air filters secured to a holding frame assembly of an air duct or air handlers of an air handling system. The final air filters and prefilters are configured to remove particles from the airstream passing through the air handling systems, such as heating, ventilation and air conditioning system.

(3) Each of the pre- and primary air filters include a filter frame into which filter media is potted, typically in an adhesive. The filter media may be configured as a pleated panel, minipleat, bag filter or arranged in V-bank of pleated panels assembled in a common filter frame. The V-bank air filters are commonly used in air filtration applications where large amounts of filter media are desirable, particularly in high flow, low pressure drop applications. Panel air filters may be used as prefilters or in air filtration applications where large amounts of filter media is not required. The adhesive securing the filter media to the filter frame prevents airflow between the filter media and filter frame, improving particle removal by reducing or removing the amount air by-passing the filter media in the air filter. In many applications requiring high filtration efficiency, a gasket is provided between the filter frame of the air filter and the holding frame assembly of the air filtration system in which the air filter is installed. The gasket prevents airflow between the air filter and holding frame assembly, improving particle removal by reducing or removing the amount air by-passing the air filter.

(4) Conventionally, the gasket permanently is attached into the filter frame. The gasket may be poured-in-place on the filter frame, secured to the filter frame with a pressure sensitive adhesive, molded into, extruded with or otherwise permanently adhered the filter frame such that replacement of the air filter or gasket necessarily requires the replacement of the entire air filter, or for the gasket to be scraped off and the filter frame prepared for another application of the adhesive. On one hand, the service life of the air filter is often much shorter than the service life of the gasket. On the other hand, the gasket may become damaged prior to the expiration of the life of the air filter. In either case, the entire air filter must be replaced when only either the filter media or gasket requires replacement, thus increasing cost of ownership and operation for the air filtration system. Another solution that may be used is to make the gasket permanently adhere to the holding frame. This solution will suffer the same shortcomings as discussed above for permanently adhering the gasket to the filter. It may be an even more disadvantageous solution since the gasket removal will then have to be made on site and often in hard to access locations.

(5) Thus, there is a need for an improved air filter and gasket for the same.

SUMMARY OF THE INVENTION

(6) Disclosed herein is a plurality of embodiments for an air filter having a filter frame and a separate filter frame gasket. The gasket has a rectangular ring shaped body having an inner perimeter, an outer perimeter and corners. A flange extends outward from a first surface of the body extending continuously along an entire length of the body. The gasket additionally has a mating portion extending from a second surface of the body extending continuously along an entire length of the body.

(7) In another embodiment, a filter frame gasket is disclosed. The filter frame gasket has a rectangular ring shaped body. The rectangular ring shaped body has an inner perimeter, an outer perimeter and corners, wherein the corners are homogeneously fully formed without a seam or weld. A first flange extends outward from an outer perimeter of the body extending continuously along an entire length of the body.

(8) Other embodiments include an air filter. The air filter includes a filter frame holding filter media within an interior of the filter frame, and a gasket elastically secured to an exterior of the filter frame. In one example, the filter frame has a body. The body may be fabricated a unitary contiguous, single mass of material, or may be assembled from separate component. The body has

a plurality of walls coupled together to form a rectangular ring having corners. The walls have an upstream surface, a downstream surface, an outer perimeter and an inner perimeter. Filter media is disposed within the ring of the body, and is very close proximity to or seal against the inner perimeter of the walls. A gasket of single piece construction disposed continuously about the plurality of walls. Alternatively, the gasket may have a seam where ends are joined to form a ring. The gasket has a rectangular ring shaped body. The gasket is secured to the filter frame by elastic compression of the stretched ring shaped body, and/or by compression of the profile of the body within a receiving feature of the filter frame. The rectangular ring shaped body has an inner perimeter, an outer perimeter and corners. A flange extends outward from a first surface of the body extending continuously along an entire length of the body, the flange extending from the outer perimeter of the filter frame and has flange corners homogenously fully formed without a seam or weld.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) So that the manner in which the above recited features of the present invention can be understood in detail, a more particular description of the invention, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this invention and are therefore not to be considered limiting of its scope, for the invention may admit to other equally effective embodiments.
- (2) FIG. 1 is a schematic sectional view of an air handler having a holding frame assembly that includes at least one air filter installed therein, with one or more air filters mounted within the holding frame, according to one embodiment of the invention.
- (3) FIGS. 2A-2B depict perspective and front views of the holding frame of FIG. 1.
- (4) FIG. 3A is a plan view of an upstream side of an air filter having a removable gasket.
- (5) FIG. 3B is a plan view of a downstream side the air filter of FIG. 3A.
- (6) FIG. 3C is a top plan view of the gasket of FIG. 3A.
- (7) FIG. 3D is a perspective view of the gasket interfaced with the filter frame of the air filter depicted in FIG. 3A.
- (8) FIG. 4A is a partial cross-sectional profile illustrating a first example of an interface between the gasket and the filter frame of the air filter of FIG. 3A.
- (9) FIG. 4B is a partial cross-sectional profile illustrating a second example of an interface between the gasket and the filter frame of the air filter of FIG. 3A.
- (10) FIG. 4C is a partial cross-sectional profile illustrating a third example of an interface between the gasket and the filter frame of the air filter of FIG. 3A.
- (11) FIG. 4D is a partial cross-sectional profile illustrating a fourth example of an interface between the gasket and the filter frame of the air filter of FIG. 3A.
- (12) FIG. 4E is a partial cross-sectional profile illustrating a fifth example of an interface between the gasket and the filter frame of the air filter of FIG. 3A.
- (13) FIG. 4F is a partial cross-sectional profile illustrating a sixth example of an interface between the gasket and the filter frame of the air filter of FIG. 3A.
- (14) FIG. 5A is a partial cross-sectional profile of an air filter having a removable gasket stretched around a periphery of a filter frame of the air filter.
- (15) FIG. 5B is a perspective view of an example corner configuration for the outside rib of the gasket of FIG. 5A.
- (16) FIG. 5C is a perspective view of another example corner configuration for the outside rib of the gasket of FIG. 5A.

(17) FIG. 6A is a partial perspective view of an air filter having another example of a gasket stretched around a periphery of a filter frame of the air filter.

(18) FIG. 6B is a partial cross-sectional view of the gasket shown in FIG. 6A.

(19) FIG. 7 is a cross-sectional view of another example for the gasket shown in FIG. 6A.

(20) FIG. 8A is a top plan view of a portion of another example of the gasket shown in FIG. 4F.

(21) FIG. 8B is a bottom perspective view of the gasket shown in FIG. 8A.

(22) FIG. 8C is a top perspective view of the gasket of FIG. 8A.

(23) To facilitate understanding, identical reference numerals have been used, wherever possible, to designate identical elements that are common to the figures. It is contemplated that elements of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(24) Embodiments of the invention generally provide a removable gasket mating to the air filter or to the holding frame having a sealing function facing outwards or inwards respectively. In one embodiment, an air filter is provided having a removable gasket. The air filter may be configured as a prefilter, a primary filter, a bag filter, a v-bank filter, or a canister filter, among other types of filter construction. The gasket is elastically secured to an exterior of a filter frame of the air filter. The gasket is ring shaped. The gasket may be fabricated a unitary contiguous, single mass of material, or may have a single seam. The gasket is secured to the filter frame by elastic compression of the stretched ring shaped body, and/or by compression of the profile of the body within a receiving feature of the filter frame. The filter frame secures a filter media within the air filter. In many applications, the air filter is installed in a holding frame assembly when in use. The holding frame assembly is attached to a housing for an air handler or other air moving conduit. The gasket disposed on the filter frame has one or both of the following functionality: the gasket retains the air filter within the holding frame; and/or, the gasket provides a modicum of sealing between the filter frame and the holding frame assembly. In one example, the gasket is molded, co-molded, or otherwise formed as a single continuous piece of material to prevent void slits, seams, opening or other asperities that would compromise the seal along the entire periphery formed by the gasket between the filter frame when disposed in the holding frame assembly. The gasket is independent of and removable from the filter frame to allow for separate reuse and replacement of either the gasket or the filter. In one or more examples, corners of the gasket are shaped to fit in the space formed at the intersection of the corners of the filter frame and the holding frame assembly to prevent voids that air may bypass or leak past the air filter when disposed in the housing. The gasket seals and holds the filter frame in a sealing position within the holding frame assembly, which in many applications desirably eliminates the need for fasteners or other devices to retain the air filter within the holding frame assembly. Below are various embodiments for the gasket in which a number of profiles for the gasket are discussed along with their advantages. Additionally, the examples of the gasket provide for different placement of the gasket on the filter frame.

(25) For the sake of clarity, the holding frame assembly and the filter frame are separate and distinct components. The holding frame assembly is part of the air handler that is permanently installed in a facility, and is not part of the air filter. Once the holding frame assembly is installed in the facility, the holding frame becomes a permanent part of the air handler. Conversely, the filter frame is part of the air filter that is manufactured, shipped and sold as a separate unit from the holding frame assembly. The filter frame is the portion of the air filter that contacts the holding frame assembly when the air filter is removably installed in the holding frame assembly. Although some filter frames have filter media elements that are removable therefrom, such filter frames still comprise the air filter itself, rather than a part of the facility once installed.

(26) Additionally, although the adjective “air” is utilized to describe the filters and filtration equipment disclosed herein, “air” is intended to mean any predominantly gas phase fluid, such as for example, combustion gases, which normally would not be described as “air”.

(27) FIG. 1 is a schematic sectional view of an air handler **100** having a holding frame assembly

102 installed therein, according to one embodiment of the invention. Although the holding frame assembly **102** is shown mounted inside the air handler **100**, it is contemplated that the holding frame assembly **102** may be utilized advantageously in or associated with other air moving conduits and ducts. Although it is contemplated the holding frame assembly **102** may be mounted in the air handler **100**, it should be appreciated that the holding frame assembly **102** may be outside the air handler **100**. For example, on the outside of a building associated with an air intake for the building there may be a filter holding frame while a fan may be on the other side of the building. (28) For the sake of simplicity, the disclosure will be discussed with the air handler **100** generally including a housing **104** that contains a blower **106** and the holding frame assembly **102**. A filter bank **108** having one or more primary air filters **110** and optionally, one or more prefilters **160** (shown in phantom) are removably mounted to the holding frame assembly **102**. The housing **104** may include an access door **122** to facilitate replacement of the primary air filters **110** and/or prefilters **160**. The blower **106** draws air into the housing **104** through a grate **112** and through the filter bank **108**, exhausting the filtered air from the housing **104** into a conduit **114**. The filtered air leaving the air handler **100** through the conduit **114** is typically routed to a facility or other location. Although the blower **106** is shown on the downstream side of the holding frame assembly **102**, the blower **106** may alternatively be located in an upstream location.

(29) The holding frame assembly **102** is sealingly coupled to the housing **104** and supports the one or more primary air filters **110** and optional prefilters **160** comprising the filter bank **108**. The holding frame assembly **102** may be sealed to the housing **104** by welding, caulking, gaskets, or other suitable methods. Generally, the holding frame assembly **102** includes one or more holding frames **116**. The holding frames **116** each having a central aperture **118** and configured to removably hold a single primary air filter **110** and/or associated prefilter **160**. The holding frames **116** are fastened, welded or otherwise permanently attached to each other such that the plurality of holding frames **116** permanently comprise part of the filter holding frame assembly **102**, and consequently, permanently comprise part of the housing **104** of the air handler **100**.

(30) The primary air filter **110** may be configured as a bag filter, a v-bank filter, or a canister filter, a pleated flat panel filter, box filter among other types of filter construction. Although the gasket below is described as utilized with the primary air filter **110**, the pre-filter **160** may similarly be configured with a gasket in the same manner for the same purpose. The primary air filter **110** includes filter media **190** disposed in a filter frame **180**. The filter media **190** may be a particulate and/or molecular phase (i.e., chemical) filter. In one example, the filter media **190** is has a Minimum Efficiency Reporting Value (MERV) rating of 7 or higher. The filter media **190** may be pleated to increase the surface area available for air passage and increase filtration efficiency of the filter element. For example, the filter media **190** is pleated and has a MERV rating of 7 or higher. In another example, the filter media **190** is a carbon-based chemical filtration media. The filter media **190** directly secured within the filter frame **180** of the primary air filter **110**. The filter media **190** may be captured within the geometry of filter frame **180** upon assembly, and/or secured by an adhesive, such as polyurethane to provide an air seal between the filter media **190** and filter frame. The filter frame **180** is comprised of four sides enclosing an open passage in which the filter media **190** is secured. The sides of the filter frame **180** may have a C-shaped profile that accepts the edge of the filter media **190** and may be fabricated from aluminum, sheet metal, plastic, paper (meaning to include paper-like products), wood or other suitable material. In the embodiment depicted in FIG. 1, the filter media has a quadrilateral plane form. In an alternative embodiment, multiple or a singular filter media element may have a single or multiple V shaped and be disposed in a single filter frame **180** to formed what is known in the air filtration industry as a V-bank air filter. In embodiments described herein, an air filter may refer to either the primary air filter **110** or the prefilter **160**.

(31) In one embodiment, primary air filters **110** may be one of American Society of Heating Refrigeration and Air Conditioning Engineers (ASHRAE) rated or a Minimum Efficiency

Reporting Value (MERV) rated filter, 95% DOP filter, a HEPA filter, or a filter of other efficiency. Other examples of primary air filters **110** that may be utilized with embodiments of the invention include, but are not limited to, 1" deep panel or pleated filters, 2" deep panel or pleated filters, 4" deep panel or pleated filters, 6" or 12" deep rigid filters, headered-style rigid or bag filters and v-bank filters, among others. In one embodiment, the prefilters **160** may be ASHRAE-rated, MERV-rated, EN779-rated, ISO-rated or other suitable first stage filters.

(32) FIGS. 2A and 2B depict a perspective view and a front plane view, respectively, of the holding frame **116** according to one embodiment of the invention. The holding frame **116** includes four I-shaped walls, e.g., a top wall **202**, two sidewalls **204**, and a bottom wall **206** that are coupled to form a rectangular ring bounding the central aperture **118**. In one embodiment, the top wall **202**, sidewalls **204**, **208** and bottom wall **206** are spaced sufficiently to accept the primary air filter **110** therebetween. The walls **202**, **204**, **206**, **208** also include back flanges **216**, **226**, **236**, **246** that extend inward 90 degrees from the plane of the walls **202**, **204**, **206**, **208** and terminate at open edges **212**, **222**, **232**, **242** that define the central aperture **118** of the holding frame **116**. The walls **202**, **204**, **206**, **208** and back flanges **216**, **226**, **236**, **246** may be fabricated from a single or multiple elements. The back flanges **216**, **226**, **236**, **246** provides a surface for the primary air filter **110** to abut when installed into the holding frame **110**, while aperture **118** defined by the open edges **212**, **222**, **232**, **242** is sized to prevent the filter **110** from passing through the holding frame **116** while not interrupting air flow through the air filter **110**. It should be noted that other angles than 90 degrees between the walls and back flanges may be used and the wall and back flanges may have slightly different shapes as long as they form a sealable and secure mount for the air filters. Additionally, it should be appreciated that the air filter **110** may be disposed on either an upstream or downstream face of the back flanges **216**, **226**, **236**, **246**.

(33) For the sake of clarity, embodiments of the invention are described in relation to a direction of airflow, for example, such as the air flow shown in FIG. 1. Further, the terms "upstream" and "downstream" may be used herein to describe the arrangement and configuration of components of embodiments of the invention relative to such a directional airflow (e.g., "downstream side"). However, it is contemplated that embodiments of the invention may be configured to be suitable for use with a "reversed" or opposite direction of airflow. As such, it is understood that such terms as "downstream" or "upstream" do not in any way limit or restrict embodiments of the invention or manners in which embodiments of the invention may be utilized, but rather, provide clarification for the arrangement and configuration of said components relative to one exemplary direction of air flow.

(34) FIGS. 3A through 3D detail an air filter, such primary air filter **110** among others, having a replaceable gasket **300**. For example, the gasket **300** may be separately attached to the filter frame **110**. The gasket **300** may be glued, over-molded, or otherwise attached to the filter frame **110** in a semi-permanent or permanent manner. Alternately, the gasket **300** may be attached to the filter frame **110** in an elastomeric fashion or with attachment features allowing the gasket **300** to be removed from the filter frame **100**. For example, a damaged gasket **300** may be removed from the filter frame and replaced with a new gasket **300**. The filter frame **110** is rectangular and may have a length of between about 300 mm to about 610 mm and even over about 1200 mm on any one side. For example, the filter frame **110** may be about 300 mm by about 300 mm, about 1 ft by about 1 ft, about 1 ft. by about 2 ft., about 2 ft. by about 2 ft. and even about 4 ft. by about 4 ft. However, it should be appreciated that the filter frame **110** may be any suitable large size for an air handling system. Herein after, the primary air filter **110** is referred to as air filter **110**. Each figure will be introduced individually for better clarity. FIG. 3A is a plan view of an upstream side of the air filter **110** illustrating the filter frame **180** and filter media **190**. Note that the air filter **110** is orientated 90 degrees relative to the illustration of FIG. 1. The filter frame **180** has a body **117**. The walls **304** of the body **117** has an inner surface **341** and an outer perimeter **342**. The walls **304** of the body **117** may have four walls, such as a first wall **381**, a second wall **382**, a third wall **383** and a fourth wall

384. The walls **304** of the body **117** meet at a respective outer-most corner **349** along the outer perimeter **342** of the body **117**. The outer-most corners **349** define the outer-most extent of the filter frame **180**. In some examples, the filter frame **180** has recesses **310** disposed at the outer-most corners **349**.

(35) FIG. 3B is a top plan view of the air filter **110** of FIG. 3A. Referring to both FIGS. 3A and 3B, the body **117** of the filter frame **180** additionally has a downstream surface **321** and an upstream surface **322**. The filter media **190** is disposed between the upstream surface **322** and the downstream surface **321**. The gasket **300** is removably disposed on the filter frame **180**. By removably disposed, it is meant that the gasket **300** is secured to the filter frame **180** without the use of adhesives. In particular, the gasket **300** is elastically secured to the filter frame **180**. By elastically secured, it is meant that the gasket **300** is secured to the holding by elastic compression of the profile of gasket **300**, and/or by compression of the ring shape form of the gasket **300** stretched around the perimeter of the filter frame **180**. The gasket **300** may be disposed on the downstream surface **321** or around the outer perimeter **342** of the filter frame **180**. When installed, the perimeter of the gasket **300** is greater than the perimeter of the filter frame **180**. The gasket **300** has an outer edge **343** that extends beyond the outer perimeter **342** of the body **117** of the filter frame **180**. In this manner, the gasket **300** forms a seal between the filter frame **180** of the air filter **110** and the holding frame **116** of the holding frame assembly **102**. Thus when the air filter **110** is installed in the holding frame **116**, air passes through the filter media **190** of the air filter **110** instead of by-passing the air filter **110** through a gap formed between the air filter **110** and the holding frame **116**.

(36) FIG. 3C illustrates a top plan view of one example of the gasket **300** suitable for use with the filter frame **180** of the air filter **110**. The gasket **300** may be configured in a number of different profiles, some of which are described below with reference to FIGS. 4A through 4F, FIGS. 5A through 5C, FIGS. 6A through 6B, and additionally in FIG. 7. The gasket **300** has a top surface **351**, an outer perimeter **343**, an inner perimeter **345** and a bottom surface (shown in FIG. 3D using reference numeral **352**). When installed to the filter frame **180**, the inner perimeter **345** faces the filter media **190**, while the outer perimeter **343** extends beyond the outer perimeter **342** of the holding frame **116**. In one example, the inner perimeter **345** of the gasket **300** is at least 1200 mm. The gasket **300** is formed from an elastomeric material. Suitable elastomeric materials include high-density foam rubber and similar materials. In another example, the gasket **300** is formed from an elastomeric material permitting the inner perimeter **345** of the gasket **300** to expand to at least 1200 mm.

(37) FIG. 3D is a perspective view of one example of an interface between the gasket **300** and the filter frame **180** of the air filter **110**. The filter media **190** is not shown in FIG. 3D, but is present and surrounded by the filter frame **180** of the air filter **110**. In some examples, the gasket **300** has a mating portion **353** along the bottom surface **352**. The mating portion **353** extends around the entire perimeter of the gasket **300**. The mating portion **353** has an inner wall, and outer wall, and a bottom. The gasket **300** also has corners **370** where each of the sides meet, thus forming the rectangular ring shape of the gasket **300**. The corners **370** are illustrated in the figures with a line where the top surface **351** of the gasket **300** changes direction to follow the walls **304** of the filter frame **180**. However, it should be understood that the line is not a seam but a change in angle at the corners **370**. That is, the corners **370** are continuous of homogenous material and not welded, glued or otherwise seamed together. Inside of the corners **370** of the gasket **300**, a key **350** may be formed along the bottom surface **352** and/or the inner perimeter **345**, and project into the interior region circumscribed by the gasket **300**. Although the key **350** is shown as a rectangle it should be appreciated the key may be oval or any other suitable shape. The mating portion **353** meets at the key **350** in the corners **370** of the gasket **300**.

(38) In one example, the gasket **300** is formed continuously without joints or seams. For example, the corner **370** of the gasket **300** is not welded or otherwise discontinuous so that air leakage is

substantially prevented. Furthermore, the gasket **300**, when formed as a single one piece seamless member, small radius, a sharp corners, or other profiles may be utilized to maximize contact between the gasket **300** and the holding frame assembly **102** for forming substantially airtight seal along the outer corners **349** of the filter frame **180**. The gasket **300** is continuous with no joints or seams and provides sealing around entire filter frame **180**. In an alternative example, the gasket **300** may include joints or seams, but not at the outer corners **349**, and may be made from multiple segments. In other examples, the gasket **300** may include joints or seams at the outer corners **349**, although such configurations increase the risk of leakage.

(39) The filter frame **180** has a channel **253**, or slot, in which the mating portion **353** of the gasket **300** is configured to interface. The key **350** of the gasket **300** is configured to interface and mate with the recesses **310** of the filter frame **180**. The recesses **310** of the filter frame **180** may be any shape, such as cylindrical or rectangular, and provide a location points for the key **350** to fix and orientate the gasket **300** to the filter frame **180**. The key **350** and mating recess **310** also help prevent the gasket **300** from rotating on the filter **110**, which may create particulate and damage the gasket **300**, but also prevent the gasket **300** from being installed in a backwards directions, which may promote leakage and/or allow the filter **110** to undesirably move within the filter frame **180**. The mating portion **353** and the key **350** have a different cross-section profile, i.e., the key **350** has a larger cross-section profile than the mating portion **353**. The different cross-sectional profile ensures the key **350** only fits in the recess **310** and preventing the key **350** from extending into the channel **253**.

(40) The key **350** of the gasket **300** may have a length and width substantially similar to that of the recess **310** so as to form a tight fit between the key **350** and the recess **310**. Similarly, the channel **253** may be substantially the same size and shape as the mating portion **353** such that a tight fit is formed between the mating portion **353** and the channel **253** when the mating portion **353** is inserted in the channel **253**. It should be appreciated, when the mating portion **353** is inserted in the channel **253** and the key **350** is inserted in the recess **310**, the gasket **300** is held firmly in place on the filter frame **180**. With the gasket **300** firmly in place on the filter frame **180**, a substantially airtight seal is formed between the gasket **300** and the filter frame **180**. Advantageously, the gasket **300** can be easily removed from the filter frame **180**, and replaced or even reused as desired. However, it should also be appreciated that the gasket **300** described herein may be glued to the filter frame **180** with a low strength adhesive. Such a gasket **300** may optionally forgo the mating portion **353** and key **350** for retaining the gasket **300** in the filter frame **180**.

(41) As illustrated in FIG. 3D, the gasket **300** is disposed on the downstream surface **321** of the filter frame **180**. The channel **253** disposed on the downstream surface **321** of the filter frame **180**. In other examples, the gasket **300** is disposed on the upstream surface **322** of the filter frame **180**. The channel **253** disposed on the upstream surface **322** of the filter frame **180**. It is contemplated that the filter frame **180** may have more than one channels **253** for example disposed on both the upstream surface **322** and the downstream surface **321** of the filter frame **180**. It should also be appreciated that the gasket **300** may alternately be stretched around the outer perimeter **342** of the filter frame **180**, such that the elastic compression of the gasket **300** holds the gasket **300** in place on the outer perimeter **342** of the filter frame **180**. In either configurations, the gasket outer edge **343** extends beyond the outer perimeter **342** of the filter frame **180** for forming a seal in the holding frame assembly **102**. Here, the channel **253** may optionally be disposed on the outer perimeter **342** of the filter frame **180**. In one embodiment the filter frame **180** has channels **253** disposed on both the upstream surface **322** and the downstream surface **321** as well as the outer perimeter **342** of the filter frame **180**. Advantageously, the filter frame **180** may have channels **253** mirrored on both sides of the filter frame **180** which permits the gasket **300** to be installed on either side.

(42) The gasket **300** may have a variety of alternative profiles, some of which are shown in FIGS. 4A-5A. FIG. 4A illustrates a first cross-sectional profile of the gasket **300**. The gasket **300** includes a ring shaped body **410**. The body **410**, shown in cross-sectional profile, has a rectangular mating

portion **353** configured to fit in the channel **253** of the filter frame **180**. Extending above the downstream surface **321** of the filter frame **180**, a first angled surface **401** extends from inner perimeter **345** to a peak **303**. The peak **303** is the most distal extent of the body **410** above the downstream surface **321** of the filter frame **180**. The peak is configured to extend about 0.1 inches to about 0.2 inches about the filter frame **180**. A second angled surface **402** extends from the peak **303** to a flat surface extending to the outer perimeter of the filter frame **180**. A flange **490** of the body **410** extends downward to the outer perimeter **343** of the gasket **300** from the second angled surface **402**. The flange **490** may have a length of 0.5 in or more. The flange **490** extends outward from the outer perimeter **342** to form a sealing feature. The flange **490** has a thickness between 0.025 inches and about 0.100 inches and has a length of between about 0.3 inches and about 1.00 inches. The flange **490** is configured to be angled between 45 degrees and 90 degrees from the outer perimeter **343** of the perimeter of the filter frame **180**, such as between about 70 degrees and about 80 degrees. The angle of that flange relative to the body, i.e., down and away from top surface, is longer than the body is wide to promote flexibility which promotes sealing and provides more forgiving wider tolerance. In one embodiment, the flange **490** may have one or more barbs **491**, a flap, serrations or other features extending therefrom. The barbs **491** aids in engaging the holding frame **116** and forming the seal when the flange **490** is compressed and or bent as the filter frame **180** is installed in the holding frame **116**. Advantageously, the gasket **300** has two sealing surfaces with one formed at the peak **303** and the other formed at the outer perimeter **343**, or flange **490**, for making a substantially air airtight seal. The gasket **300** can be made fairly rigid and solid for longevity, while minimizing excess material.

(43) FIG. 4B illustrates an alternative cross-sectional profile of the gasket **300**. The gasket **300** has ring shaped body **420**. The body **420** has a flange **490** that is substantially similar to that discussed with respect to FIG. 4A. The body **420** has a rectangular mating portion **353** configured to fit in the channel **253** of the filter frame **180**. The body **420** has an inner flat surface **492** that helps to stabilize the gasket **300** on the downstream surface **321** of the filter frame **180**. Extending above the downstream surface **321** of the filter frame **180** is a portion of the body **420** that is substantially rectangular in shape. The body **420** has a top surface **429** that forms the sealing surface. The top surface **429** of the body **420** is substantially flat. The top surface **429** the body **420** may be parallel to the downstream surface **321** of the filter frame **180**. Advantageously, the gasket **300** has two sealing surfaces, one at the top surface **429** the body **420** and other at the outer perimeter **343** for making an air airtight seal. The gasket **300** can be made rigid and solid for longevity to promote reuse.

(44) FIG. 4C illustrates another alternative cross-sectional profile of the gasket **300**. The gasket **300** has a body **430**. The body **430** has a rectangular mating portion **353** configured to fit in the channel **253** of the filter frame **180**. The mating portion **353** may extend between about 0.15 inches to about 0.5 inches into the channel **254**. The body **430** has an inner flat surface **492** that helps to stabilize the gasket **300** on the downstream surface **321** of the filter frame **180**. The body **430** has a flange **490** that is substantially similar to that discussed with respect to FIG. 4A. The body **430** has a rounded top surface **439**. The rounded top surface **439** extends from the inner flat surface **492** to the flange **490**. The body **430** has a hollow **311** disposed below the rounded top surface **439** and above the rectangular mating portion **353**. The hollow **311** is fully enclosed to an interior of the body **430**. The hollow **311** allows the rounded top surface **439** of the body **430** to be compressed for easier installation of the air filter **110** into the holding frame assembly **102**, while forming a good seal between the filter frame **180** and the holding frame **116**. Advantageously, the gasket **300** has two sealing surfaces, one at the rounded top surface **439** the body **430** and other at the outer perimeter **343** of the gasket **300** for making a substantially air airtight seal. The flange **490** also provides stability and holds the filter into the holding frame **116**.

(45) FIG. 4D illustrates another alternative cross-sectional profile for the gasket **300**. The gasket **300** has a ring shaped body **440**. The body **440** has a rectangular mating portion **353** configured to

fit in the channel 253 of the filter frame 180. The body 440 has a flange 490 that is substantially similar to that discussed with respect to FIG. 4A. The body 440 has an inner flat surface 492 that helps to stabilize the gasket 300 on the downstream surface 321 of the filter frame 180. The body 440 has a rounded top surface 449. The rounded top surface 449 extends from the flange 490 toward the inner flat surface 492. A gap 441 is provided between the rounded top surface 449 and the inner flat surface 492 such that the rounded top surface 449 does not extend all the way to the inner flat surface 492. The body 440 has a hollow 311 disposed in an interior of the body 440 below the rounded top surface 449 and above the rectangular mating portion 353. The hollow 311 is exposed to an exterior portion of the body 430 through the gap 441. The hollow 311 allows the rounded top surface 449 of the body 430 to be compressed for easier installation of the filter frame 180 and forming a good seal. The gap 441 allows air to escape the hollow 311 upon installation of the air filter 110 in the holding frame 116, making for a more compliant and robust seal.

Advantageously, the gasket 300 has two sealing surfaces, one at the rounded top surface 449 of the body 430 and the other at the outer perimeter 343 or the flange 490.

(46) FIG. 4E illustrates another cross-sectional profile for the gasket 300. The gasket 300 has a ring shaped body 450. The body 450 has a rectangular mating portion 353 configured to fit in the channel 253 of the filter frame 180. The body 450 has a flange 490 that is substantially similar to that discussed with respect to FIG. 4A. The body 450 has an inner flat surface 492 that helps to stabilize the gasket 300 on the downstream surface 321 of the filter frame 180. The body 450 has a rounded top surface 459. The rounded top surface 459 extends fully from the flange 490 to inner perimeter 345 of the gasket 300. The body 450 is solid throughout. Advantageously, the gasket 300 has two sealing surfaces, one at the rounded top surface 459 the body 430 and the other at the outer perimeter 343 and/or the flange 490.

(47) The material and profile selection for the gasket 300 facilitates deflection of the gasket 300 during the installation of the air filter 110 into the holding frame 116 and compression of the gasket 300 for securing the filter 110 into the holding frame assembly 102, while requiring no secondary mechanical fasteners. The gasket 300 deflects during filter installation to seal the air filter 110 to the holding frame assembly 102. Additionally the gasket 300 forms a continuous ring around the entire the outer perimeter 343 of the perimeter of the filter frame 180, including the outer corners 349, so that potential air by-pass is minimized.

(48) FIG. 4F is a partial cross-sectional profile illustrating a sixth example of an interface between the gasket 300 and the filter frame 180 of the air filter 110 of FIG. 3A. The gasket 300 has a ring shaped body 460. The body 460 has a rectangular mating portion 353 configured to fit in the channel 253 of the filter frame 180. The body 460 has a flange 490 that is substantially similar to that discussed with respect to FIG. 4A. The body 460 has an inner flat surface 492 along an interior edge 465 of the gasket 300 that helps to stabilize the gasket 300 on the downstream surface 321 of the filter frame 180. The body 460 has a flat top surface 469. The flat top surface 459 has a protrusion 464 extending from the flat top surface 459 at or near the flange 490. The flat top surface 469 extends from the protrusion 464 to the interior edge 465 of the gasket 300. The protrusion 464 extends away from the flat top surface 469 at an angle 466 towards the interior edge 465. The angle 466 of the protrusion 464 determines a height 467 the protrusion 464 extends above the flat top surface 469. The angle 466 of the protrusion 464 may be between 90 degrees and about 30 degrees such as 60 degrees.

(49) The body 460 is formed by a plurality of elements arranged in a ladder type fashion in the rectangular mating portion 253. The rectangular mating portion 253 has an outer wall 462 connected to an inner wall 461 by a plurality of cross members 463 (the front of one of which is shown in the sectional view of FIG. 4F). The plurality of cross members 463 may be substantially orthogonal to the outer wall 462 and the flat top surface 469. The plurality of cross members 463 may additionally be substantially orthogonal to the inner wall 461. The plurality of cross members 463 may be spaced apart to form hollows therebetween or a ladder structure in the rectangular

mating portion **253**. Alternately, the cross members **463** may have any suitable geometry, such as that of a honeycomb structure below the flat top surface **469** and forming the rectangular mating portion **253**, that allows the minimization of the amount of material used to reach maximum stiffness with minimal weight and material cost. In one example, the outer wall **462**, the plurality of cross members **463**, and the inner wall **461** form a ladder structure in the bottom of the rectangular mating portion **253**. The ladder structure also depicted in FIG. **8B**.

(50) Continuing to refer to FIG. **4F**, the gasket **300** advantageously has two sealing surfaces, one at the protrusion **464** and one at the flange **490**. The material and profile selection for the gasket **300** facilitates deflection of the gasket **300** during the installation of the air filter **110** into the holding frame **116** and compression of the gasket **300** for securing the filter **110** into the holding frame assembly **102**, while requiring no secondary mechanical fasteners. Additionally, the array of elements forming the mating portion **253** reduces the of material resulting in a cost savings while additionally allowing the mating portion **253** to be compressed for easy installation. The gasket **300** deflects during filter installation to seal the air filter **110** to the holding frame assembly **102**. Additionally the gasket **300** forms a continuous ring around the entire the outer perimeter **343** of the perimeter of the filter frame **180**, including the outer corners **349**, so that potential air by-pass is minimized.

(51) FIG. **5A** illustrates yet another cross-sectional profile for the gasket **300**. The gasket **300** has a ring shaped body **510** having an outwardly extending flange **490**. The gasket **300** has an outside rib **501** extending from a bottom of the flange **490** immediately adjacent to the flat surface of the body **510**. The downstream surface **321** of the filter frame **180** is configured to insert into a slot formed between the body **510** and the outside rib **501** to assist securing the gasket **300** to the filter frame **180**. Although gasket **300** is shown in cross-section to be substantially similar to that of gasket **300** shown in FIG. **4A**, it should be appreciated that the gasket **300** illustrated in FIG. **5A** may have other cross-sectional profiles, such as but not limited to as shown in FIGS. **4A** through **4F**. The gasket profiles illustrated in FIGS. **4A** through **4F** may optionally include an outside rib **501**. The outside rib **501** is configured to prevent the gasket **300** from rolling out or off of the filter frame **180** and help secure the gasket in place while there is a moment force applied to the flange **490** during installation of the filter frame **180**.

(52) The outside rib **501** is spaced apart from the rectangular mating portion **353** such that a portion of the filter frame **180** is sandwiched between them. The outside rib **501** may be in direct contact with the filter frame **180** when the rectangular mating portion **353** is in the channel **253** of the filter frame **180**. The outside rib **501** may be rectangular in shape, oblong, triangular or any shape such that a flat of the outside rib **501** is adjacent the outer perimeter **342** of the filter frame **180**. The outside rib **501** may be continuous around the length of the gasket **300**. Alternately, the outside rib **501** may be formed of segments **571**, **572**, such as four or more segments. FIGS. **5B** and **5C** illustrate the outside rib in the corner section that may not necessarily be indicative of the continuous nature of the outside rib **501** as the outside rib **501** in some examples may only be in the corner for holding the gasket **300** firmly in place in the filter frame **180**. In some examples having the outside rib **501** in the corners **370**, the gasket **300** may forgo the inclusion of the key **350** for holding the gasket **300** in place in the filter frame **180**.

(53) FIG. **5B** illustrates a perspective view of an example corner configuration for the outside rib **501** of the gasket **300** of FIG. **5A**. The outside rib **501** does not extend completely to the corner **370** of the gasket **300**. The outside rib **501** may extend to a distance **560** from the outer corner **349** of the filter frame **180**. The outer corner **349** additionally aligns with the corner **370** of the gasket **300**. The distance **560** may be between about 1 mm to about 50 mm or more. The outside rib **501** having the distance **560** from the outer corner **349** of the holding frame **116** permits the gasket **300** to be installed more easily while still having the benefit of the outside rib **501** for stabilizing and fixing the gasket **300** to the filter frame **180**.

(54) FIG. **5C** illustrates a perspective view of another example corner configuration for the outside

rib **501** of the gasket **300** of FIG. 5A. The outside rib **501** extends completely to the corner **370** of the gasket **300** and likewise the outer corner **349** of the filter frame **180**. Thus, a rib corner **515** is formed where the outside rib **501** on adjacent sides, i.e., walls **202**, **204**, **206**, **208**, of the filter frame **180** meet. The rib corner **515** is continuous of homogenous material and not welded, glued or otherwise seamed together. The outside rib **501** permits the gasket **300** to be tightly secured to the filter frame **180**. Additionally, the outside rib **501** having the rib corner **515** is less prone to damage when the gasket **300** is being installed on the filter frame **180**.

(55) It should be appreciated that the gasket, substantially similar in purpose to gasket **300**, may also reside on the outer perimeter **342** of the filter frame **180**. FIG. 6A illustrates a perspective view of an example gasket **600** disposed on the outer perimeter **342** of the filter frame **180**. FIG. 6B illustrates a cross-sectional view of one example of the gasket **600** shown in FIG. 6A. The gasket **600** has a T-shaped profile and a ring shaped body **610** that is stretched over the filter frame **180** so the elastic compression of the ring retains the gasket **300** on the exterior perimeter of the filter frame **180**. The T-shaped body **610** has a central flange **690** extending from a base portion **612** of the body **610**. It should be appreciated that the central flange **690** may bisect the base portion **612** of the body **610**. It should also be appreciated that the central flange **690** may be offset from a centerline bisecting the base portion **612** of the body **610**. The base portion **612** is configured to be disposed on the outer perimeter **342** of the filter frame **180**. For example, a distance **691** from a first end **601** to a centerline bisecting the central flange **690** may be more or less than a distance from the centerline to a second end **602** of the base portion. A distal end **643** of the central flange **690** forms the sealing surface between the filter frame **180** and the holding frame **116**.

(56) The gasket **600** may be formed as described above with reference to the gasket **300**. A corner **675** of the gasket **600** is formed continuously with the adjoining sides of the gasket **600**. The corner **675** of the gasket **600** may be formed to have a radius, a sharp corner, or other profile to maximize contact between the gasket **600** and the holding frame **116** for forming an airtight seal along the outer corners **349** of the filter frame **180**.

(57) In some examples the gasket **600** may sit in a channel (not shown) formed in the walls **304** of the filter frame **180**. In other examples, the gasket **600** is disposed directly on the flat of the walls **304**. The gasket **600** may have a width in the base portion **612**, i.e., at least twice the distance **691**, sufficient to prevent rolling and maintain good frictional adhesion to the walls **304** such that the gasket **600** does not slide off the walls **304** when installing the filter frame **180** into the holding frame **116**. Advantageously, gasket **600** can be adapted to existing filter frames while providing a good seal between the filter frames and holding frames.

(58) As discussed above, optional channel **253** may be disposed on the outer perimeter **342** of the filter frame **180** for locking the gasket **700** to the filter frame **180** and preventing it from sliding while being the filter frame **180** is installed in the holding frame **116**. In one example, the channel **253** may be configured to fit the base portion **612** of the gasket **700**. In another example, the channel **253** may be configured to fit a protrusion extending inward from the base portion **612** of the gasket **700**.

(59) FIG. 7 illustrates an orthogonal view of another example gasket **700** that may be utilized in place of the gasket **600** shown in FIG. 6A. The gasket **700** is substantially similar to gasket **600** with the exception of the replacement of the central flange **690** with a plurality of flanges. In the example depicted in FIG. 7, the gasket **700** includes a first flange **781** and a second flange **782**. The gasket **700** may alternatively include additional flanges similar to the flanges **781**, **782**. The first flange **781** is substantially similar to the second flange **782**. The first flange **781** and second flange **782** have corner **795** that may be formed at a radius or a 90° angle. The corner **795** is shaped in a manner to provide a good seal between the outer corner **349** of the filter frame **180** and the holding frame **116**. Advantageously, the first flange **781** and the second flange **782** provide a redundancy in the seal to prevent leaks or bypass of air in the gasket **700**. Additionally, the first flange **781** and second flange **782** balance the gasket **700** to further minimize rolling or movement of the gasket

700 when the filter frame 180 is installed in the holding frame 116. Furthermore, the multiple flanges 781, 782 advantageously increases the frictional holding force that retains the air filter 110 within the holding frame 116.

(60) FIG. 8A is a top plan view of a portion of another example of the gasket 800 shown in FIG. 4F. The gasket 800 is similar to gasket 300 shown in FIGS. 3A-3D. The flange 490 of the gasket 800 has cutaways 854 at a corner 801 of the gasket. The flange 490 has cutaways 854 adjacent to each side of each corner 801. The gasket 800 has a cross-sectional profile illustrated in FIG. 4F and additionally has a uniquely different corner profile on the flange 490 as shown in FIGS. 8A-8C. However, it should be appreciated that may be configured in a number of different profiles, some of which were described with reference to FIGS. 4A through 4F. For clarity of understanding, FIG. 8A will be discussed in conjunction with FIG. 8C. FIG. 8C is a top perspective view of the gasket of FIG. 8A.

(61) The gasket 800, similar to gasket 300, is formed in a continuous rectangular ring shape body 460 from an elastomeric material. The gasket 800 is suitable for use with the filter frame 180 of the air filter 110. In one example, the inner perimeter 345 of the gasket 300 is at least 1200 mm. In another example, the gasket 300 is formed from an elastomeric material permitting the inner perimeter 345 of the gasket 300 to expand to at least 1200 mm.

(62) The gasket 800 has a top surface 351, outer perimeter 343, inner perimeter 345 and bottom surface 352 similar to gasket 300. The flange 490 extends outwardly from the body 460 forming the outer perimeter 343. An outer edge 807 of the flange 490 extends along the outer perimeter 343. When installed to the filter frame 180, the inner perimeter 345 faces the filter media 190, while the outer perimeter 343 extends beyond the outer perimeter 342 of the holding frame 116. The body 460 has the protrusion 464 extending from the top surface 351 at or near the flange 490.

(63) The corners 801 are illustrated in the figures with lines where the top surface 351 of the gasket 800 changes direction. However, it should be understood that the line is not a seam but a change in angle at the corners 801. That is, the flanges 490 are continuous of homogenous material and not welded, glued or otherwise seamed together in the corners 801.

(64) FIG. 8B is a bottom perspective view of the gasket 800 shown in FIG. 8A. The flange 490 has a pair of cutaways 854 at the corners 801. The cutaways 854 are immediately adjacent the corners 801. The cutaways 854 in the flange 490 may linearly extend from the corners 801 prior to transitioning to the outer edge 807. Alternately, the cutaways 854 may extend from the corners 801 in an arc, S-curve, or other shape such that the cutaways 854 at the corner 801 has a first extension 883/884 disposed inward a distance 885 of a second extension 881/882 of the outer edges 807. The cutaways 854 permit the flange 490 of the gasket 800 to fold in the corners 801 and form a 90 degree sealing surface preventing leaks.

(65) In one example, the gasket 800 additionally has a little cutaway 852 in the flange 490. The little cutaway 852 may provide clearance for the gasket 800 when installing in the frame. The little cutaway 852 may be disposed in one or more locations on the gasket 800. In one example, the little cutaway 852 is disposed between one of the cutaways 854 and an adjoining outer edge 807.

(66) This has the effect of creating a longer sealing edge from one corner 801 to an adjacent corner 801. For example, the length from one corner 801 to an adjacent corner 801 is shorter than the length of the combined outer edge 807 and cutaway 854. Additionally, the cutaways 854 reduce the amount of material in the corner 801. Advantageously, the cutaways 854 in the gasket 800 prevents a fold of material in or around the corner 801 for creating a better seal while increasing the length of the seal.

(67) The body 460 has the rectangular mating portion 353 configured to fit in the channel 253 of the filter frame 180. The gasket 800 also has a key 850 (substantially similar to key 350) formed along the bottom surface 852 and/or the inner perimeter 345. The key 850 projects into the interior region circumscribed by the gasket 800. Although the key 850 is shown as a rectangle it should be appreciated the key may be oval or any other suitable shape. A mating portion 853 (substantially

similar to mating portion 353) meets at the key 850 in the corners 801 of the gasket 800.

(68) In one example, the key 850 and the mating portion 853 have hollow portions. The key 850 and the mating portion 853 is formed in an array of elements. The outer wall 462 is connected to the inner wall 461 by the plurality of cross members 463 in the key 850. The plurality of cross members 463 may provide stiffness to the key 850 while reducing the amount of material in forming the gasket. Furthermore, the combination of cross members 463 can be sized and configured to adjust the compression of the key 850 such that key 850 is easier to install while having a better hold in mating with the recesses 310 of the filter frame 180.

(69) The examples of the gaskets that are independent of the air filter shown and discussed above provide a number of benefits over conventional gasketed filter frames. Advantageously, the continuous nature of the gasket with the full corners and arrangement of sealing surfaces provided by the gasket offers a better seal between the filter frame and the holding frame of the air handler system. The various gaskets disclosed herein may be utilized with filter frames having complimentary mating features, and additionally conventional air filter frames. The reusability of the gasket independent of the air filter, as well as the ability to separately replace the gasket and air filter reduces overall cost of maintaining the air handling system.

(70) In another embodiment, the removable gasket is disposed on the holding frame. The gasket mates with the holding frame. The holding frame gasket has a rectangular ring shaped body. The rectangular ring shaped body has an inner perimeter and an outer perimeter. A channel may be formed in the inner perimeter to hold the gasket. The channel may have one or more slots for configured to accept an aperture of the gasket. The rectangular gasket has an inner surface and an outer surface along a periphery of the gasket and opposite the inner surface. The gasket has corners wherein the corners are homogenously fully formed without a seam or weld. A first flange extends inwards from the inner surface of the rectangular ring shaped body extending continuously along an entire length of the body. The first flange has fully formed corners similar to discussed above for forming a seal with the filter frame. The gasket provides a seal for the air filter frame inserted therein.

(71) While the foregoing is directed to embodiments of the invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A replaceable air filter frame gasket, comprising: a rectangular ring shaped body having a top surface, a bottom surface, an inner wall, and an outer wall; an inner perimeter; an outer perimeter having a first corner; a flange extending outward to the outer perimeter from the outer wall of the body along the outer perimeter, wherein the flange has a first cutaway at the first corner extending from the first corner to a first outer edge of the outer perimeter and a second cutaway extending from the first corner to a second outer edge of the outer perimeter; a protrusion extending upward from the top surface; and a mating portion extending along the bottom surface of the body.
2. The replaceable air filter frame gasket of claim 1 further comprising: a rib extending from the flange in a direction of the mating portion.
3. The replaceable air filter frame gasket of claim 1, wherein the first corner of the outer perimeter is disposed a first distance inward from the first outer edge and a second distance inward from the second outer edge.
4. The replaceable air filter frame gasket of claim 3, wherein the first cutaway and the second cutaway is adjacent to each side of the first corner of the flange.
5. The replaceable air filter frame gasket of claim 1, further comprising: a key extending from the bottom surface, the key disposed proximate the first corner of the rectangular ring shaped body.
6. The replaceable air filter frame gasket of claim 5, wherein the mating portion and the key have a

different cross-section profile and the key has a larger cross-sectional profile than the mating portion.

7. The replaceable air filter frame gasket of claim 1, wherein the protrusion extends away from the top surface at an angle towards the inner perimeter.

8. The replaceable air filter frame gasket of claim 7, wherein the angle of the protrusion is between about 90 degrees and about 30 degrees.

9. The replaceable air filter frame gasket of claim 1, further comprising: a cutaway disposed in the flange.

10. An air filter comprising: a rectangular filter frame having frame corners; filter media disposed in the filter frame; and the replaceable air filter frame gasket of claim 1 elastically secured to and extend outward of an outer perimeter of the filter frame.

11. The filter frame of claim 10, wherein the rectangular filter frame further comprises: a body wherein a downstream surface of the body has a channel and a recess.

12. The filter frame of claim 11, wherein replaceable air filter frame gasket further comprises: a key configured to fit in the recess, wherein a horizontal profile of the key is shaped differently than that of the mating portion and the key does not fit in the channel.

13. The filter frame of claim 12, wherein the rectangular ring shaped body has an enclosed hollow interior.

14. The filter frame of claim 12, wherein the flange further comprises: a second corner having a third cutaway and a fourth cutaway; a third corner having a fifth cutaway and a sixth cutaway; and a fourth corner having a seventh cutaway and an eighth cutaway.

15. The filter frame of claim 14, wherein each of the first through eighth cutaway is of substantially similar shape and size.
